Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Cobalt-free and high-rate stable 5V lithium nickel manganese oxide spinel cathodes enabled via surface oxygen vacancies			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Batteries	Battery Materials

Concept of Proposed Technology

Rechargeable lithium-ion batteries (LIBs) remain the dominant choice for most energy storage applications including electric vehicles, hybrid electric vehicles, stationary storage and portable electronics. Different types of LIB cathodes have been interchangeably deployed in the industry including layered oxides (such as $LiCoO_2$ (LCO), $LiNi_xMn_yCo_{1-x-y}O_2$ (NMC) and $LiNi_xCo_yAl_zO_2$ (NCA)) that display high capacities (> 160 mAh g^{-1}) but are costlier due to inclusion of cobalt and may pose a safety risk because of gas release (O_2, CO_2, CO) at high voltage.[1] On the other hand, polyanion type phosphates such as $LiFePO_4$ show superior high-rate capability and safety, but operate at a reduced average voltage (~3.4 V) and thus exhibit lower energy density.[2]

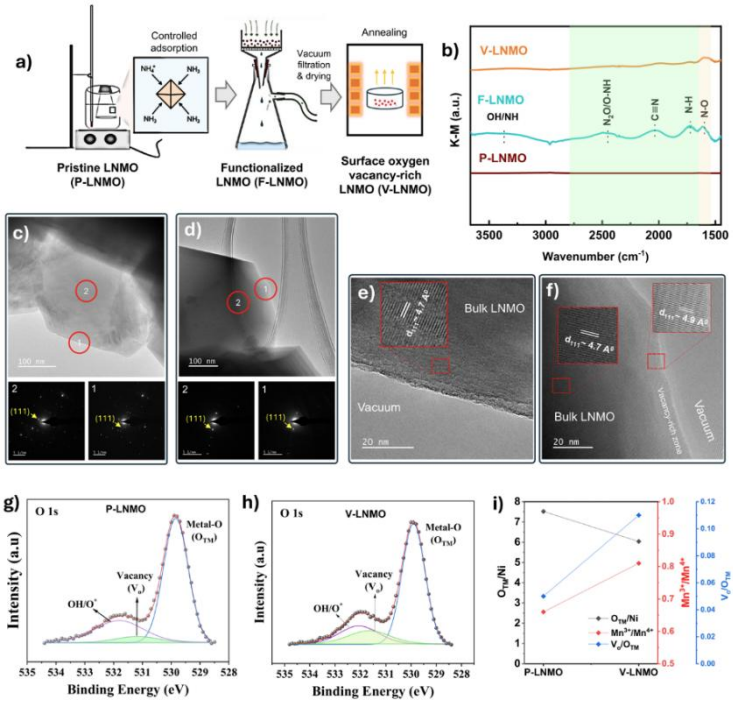
Recently, spinel $LiNi_{0.5}Mn_{1.5}O_4$ (LNMO) has emerged as a promising LIB cathode candidate combining the benefits of high energy given its high operating voltage (> 4.6 V), low-cost (Co-free), fast lithium-ion transport within its 3D channels, and good safety.[3-4] However, LNMO suffers from rapid capacity decay caused by electrolyte instability at the interface under high voltage. Electrolytes containing carbonate-based solvents (EC, EMC, DMC etc.) tend to oxidize at voltages > 4.5 V (vs. Li/Li^+) generating byproducts[5] (such as O_2, CO_2, CO, POF_3, $C_2H_5OCOOPF_4$, OPF_2ORF), which leads to formation of an inferior cathode electrolyte interphase and ultimately poor cycling stability. These parasitic reactions also irreversibly increase the acidity in the electrolyte, accelerating the dissolution of Mn^{2+} from the spinel lattice. The aforementioned reactions exacerbate at high temperatures or high rates.[5] Therefore, it is imperative to alleviate these parasitic reactions for high-rate stability in LNMO based LIBs.

To mitigate these interfacial challenges, two primary strategies have emerged – surface coating and elemental doping. Many surface coatings have been studied for LNMO, such as Al_2O_3,[6] CuO,[7] Co_2O_3,[8] SiO_2,[9] and $LiPO_4$.[10] Cationic doping in the particle bulk such as B,[11] V,[12] Ti,[13] Ge,[14] and Ga[15] have also demonstrated improvement in the cycling performance at high voltage. However, both these methods have not shown appreciable impact in improving cycling stability of LNMO cathodes at high rates (i.e., 1C or higher). Beyond these established strategies, a few recent studies have explored the role of "bulk oxygen vacancies" in altering the electronic structure and electrochemical performance of spinel cathodes.[16-17] These efforts have concentrated exclusively on *in situ* generation of bulk oxygen vacancies during high-temperature particle synthesis (by use of reducing atmospheres) with limited success in terms of performance

	improvements. In fact, in some cases, the specific capacity and cycle life degraded due to oxygen vacancies in the particle bulk.[16-17] The reasons for this are not clear but could be related to the fact that the concentration of oxygen vacancies that can be generated in the particle bulk is not large enough to significantly influence particle conductivity and the overall charge transfer resistance of the electrode. In this project, **we propose a different approach**. Instead of creating oxygen vacancies in the bulk particle, we focus exclusively on the **"near surface region"** which is in direct contact with the electrolyte. The near surface region plays a decisive role in dictating the electrode's charge transfer resistance, and hence it is crucial to establish a high concentration of oxygen vacancies in this region. We demonstrate that treatment of LNMO particles with ammoniacal solutions followed by annealing enables targeted oxygen vacancy formation, limited to the near-surface region of the particle. **In our approach, by generating a high concentration of oxygen vacancies near the particle surface, we boost the electronic conductivity of the vacancy-enriched LNMO (V-LNMO) electrode by > 3-fold compared with the pristine LNMO (P-LNMO) electrode. Moreover, we report > 2-fold increase in Li^+ diffusivity for V-LNMO as compared to P-LNMO. We also find that V-LNMO is a far weaker oxidizing agent when compared to P-LNMO, thus greatly suppressing electrolyte oxidation at high voltages.** These attributes of V-LNMO result in an across-the-board improvement in self-discharge, specific capacity, energy density, rate capability, coulombic efficiency and cycling stability. When cycled at ~200 mA g^{-1}, the capacity fade averaged over 3,000 cycles for the surface vacancy-enriched material is ~0.0167% per cycle as compared to an order of magnitude higher fade-rate for the baseline material. **The V-LNMO cathodes proposed here offer a superior alternative to existing LIB cathodes, combining the high rate, cobalt-free and low-cost advantages of LFP with the high energy density of NMC and NCA cathodes.**
Rationale for Joint R&D Project	A joint project is needed for two specific reasons: • Our testing is limited primarily to the coin-cell level since we **lack expertise for larger form factor pouch or prismatic cell testing**, which is necessary to demonstrate the proof-of-concept. **Industry is ideally suited to direct such scale-up work.** • We have conducted a preliminary cost ($/kWh) analysis. The cost per kWh is computed based on individual prices of the cathode materials ($/kg) divided by their respective energy density (kWh/kg). High energy cathodes like NMC and NCA have high cost associated with them, making each energy unit more expensive at roughly ~$40.32 and ~$46.94 per kWh, respectively. By contrast, V-LNMO and LFP are superior from the price point of view with close to ~$26.77 and ~$30.59 per kWh. Our preliminary cost estimates for V-LNMO did not include the expense associated with ammoniacal treatment and annealing to engineer the surface oxygen vacancies. **A detailed manufacturing scalability and cost analysis that models these costs explicitly will be necessary to establish the true value of this technology in terms of the $/kWh—an industrial partner is ideally suited for such an undertaking.**
Key Words	1. Environmentally friendly (cobalt-free) batteries 2. Surface oxygen vacancy engineered $LiNi_{0.5}Mn_{1.5}O_4$ material 3. Improved high voltage stability and energy density 4. Improved electronic conductivity and Li^+ diffusivity 5. Improved high-rate stability, power density and reduced cost

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	• It is important to establish the "optimal oxygen vacancy concentration" for maximum electronic and ionic conductivity increase in V-LNMO particles, as well as maximizing V-LNMO's tendency to retard electrolyte oxidation at high voltages. • We propose to regulate various synthesis parameters associated with the ammoniacal (NH_4OH) and annealing treatment to achieve the desired (optimal) oxygen vacancy concentration. Our goal will be to achieve high energy density and stable cycling performance at elevated charge-discharge rates. • In our preliminary work, we have tested V-LNMO at electrode mass loadings of 8 to 12 mg cm^{-2}. In this project, we propose to achieve industrial standard mass loadings of 20 to 30 mg cm^{-2}.			

	• We will construct practical full-cell devices with the optimized V-LNNO cathodes paired against commercial battery-grade graphite anodes. Detailed electrochemical characterization of the full cells will be pursued to establish key performance metrics. Pouch cell based minimum viable prototypes will be demonstrated. • The project will culminate with a technoeconomic analysis to assess manufacturing scalability and economic viability of V-LNMO cathodes for large-scale industrial deployment.
R&D Target Specifications	The **key deliverable** for this project is: a fully optimized V-LNMO material, which offers the best possible combination of energy and power density, high-rate performance, cycle life, reduced cost and enhanced safety. Further, a **minimum viable prototype (MVP)** full-cell device in a pouch cell form factor will be demonstrated. MVP target specifications include: • Stable performance (fade rate < 0.01% per cycle) over 3000+ cycles at C-rates of up to 2C. • The gravimetric and volumetric energy density should exceed that of LFP cathodes, while the gravimetric and volumetric power density should exceed that of NMC and NCA cathodes. • The cost should be in the range of ~30 $/kWh, i.e. comparable to LFP cathodes and much cheaper than NMC and NCA cathodes.
R&D Plan	**1. Preliminary Results** **Figure 1a** shows a schematic of the treatment process that was developed to engineer V-LNMO materials. First, P-LNMO powder was well-dispersed in a concentrated ammoniacal (NH_4OH) solution maintained at ~60 ^{0}C for ~45 minutes. The pH of this solution was measured to be ~12. At this alkaline pH, the P-LNMO surface is negatively charged (zeta potential ~ -38 mV), suggesting that positively charged ammonium (NH_4^+) ions will adsorb onto the P-LNMO particle surface, creating a functionalized particle (F-LNMO). FT-IR (Diffuse Reflectance) spectra for P-LNMO and F-LNMO powders was carried out (**Figure 1b**) to identify the surface functional groups that result from the ammoniacal treatment. F-LNMO shows a broad peak at ~3370 cm^{-1}, ~2400 cm^{-1}, ~2075 cm^{-1}, ~1700 cm^{-1} and ~1590 cm^{-1} corresponding to OH/NH stretching modes, O-NH/N_2O, C≡N stretching, N-H and N-O interaction respectively.[18] These peaks are entirely absent in P-LNMO, which confirms the successful attachment of nitrogen functional groups onto the F-LNMO surface due to the ammoniacal treatment.  **Figure 1:** *(**a**) Schematic of ammonia solution treatment process. (**b**) FTIR of starting materials (P-LNMO), nitrogen-functionalization on the surface (F-LNMO) and vacancy enriched V-LNMO. TEM micrographs of (**c**) P-LNMO and (**d**) V-LNMO along with their corresponding SAED patterns below (SAED1 - surface region, SAED2 - bulk region). High magnification micrographs of (**e**) P-LNMO and (**f**) V-LNMO highlighting different surface characteristics along with d-spacing comparison from SAED presented in the insets. High resolution XPS O1s spectra of (**g**) P-LNMO and (**h**) V-LNMO highlighting the oxygen vacancy peak (V_o). (**i**) Surface compositional analysis from the obtained XPS intensities.* Next, F-LNMO was annealed at ~500°C in ambient for ~3 hours and then allowed to cool naturally. Nitrogen related FT-IR peaks in the shaded regions (**Figure 1b**) reduce in intensity upon annealing, suggesting N-O gaseous release at high temperatures, leaving behind vacant oxygen sites (V_o) near the surface. To understand the changes in surface structure of V-LNMO relative to P-LNMO, transmission electron microscopy/selected area electron diffraction (TEM/SAED) analysis was conducted (**Figures 1c–f**). The positions of SAED are indicated by red circles in the low magnification micrographs in Figure 1c and d and their corresponding SAED patterns are presented under it (1: Surface region, 2: Bulk region). As indicated in the high magnification image for V-LNMO

in Figure 1f, the surface lattice has (111) planes coherent with the bulk, as expected for the spinel structure, however a d-spacing of 4.9 Å is observed near the surface as opposed to 4.7 Å for the bulk. Similar observations were noted for other planes. This difference likely accounts for an altered spinel structure with higher Mn^{3+}/Mn^{4+} ratio where the trivalent Mn has a "larger ionic radius" when compared to the quadrivalent Mn.[19] On the contrary, no difference in the lattice d-spacing was observed for P-LNMO (Figure 1e) when comparing the surface with the bulk. It should be noted that oxygen vacancies release excess electrons that localize on nearby Mn cations, partially converting $Mn^{4+} \rightarrow Mn^{3+}$ and thus increasing the net Mn^{3+}/Mn^{4+} ratio. Thus, the increased d-spacing observed in the near surface region for V-LNMO suggests an increased oxygen vacancy concentration in that region. To confirm this, we carried out surface composition analysis by using X-ray Photoelectron Spectroscopy (XPS). High-resolution spectra corresponding to O1s of P-LNMO and V-LNMO are shown in **Figures 1g** and **1h,** respectively. The peak at ~ 529.8 eV corresponds to oxygen bonded to the transition metal (i.e., Mn or Ni)[20] (denoted as O_{TM}). Besides this, we also observed a peak reflecting the presence of oxygen vacancy (~531.4 eV)[21] (denoted as V_o). Notably, the V_o peak is much higher in intensity for V-LNMO relative to P-LNMO, confirming an increase in surface oxygen vacancies. This contrast is further highlighted in the surface compositional analysis provided in **Figure 1i**. For V-LNMO, the peak intensity of O_{TM} normalized by the Ni concentration reduces by ~20%, indicating presence of spinel $LiNi_{0.5}Mn_{1.5}O_{4-\delta}$ phase at the surface, where $\delta_{surface} > \delta_{bulk}$. Similarly, the ratio of oxygen vacancy (V_o) concentration to O_{TM} in V-LNMO is ~2.2-fold higher than that of P-LNMO. The Mn $2p_{3/2}$ XPS spectra (not shown here) confirmed Mn^{4+} reduction as reflected in the ratio of Mn^{3+}/Mn^{4+} (**Figure 1i**), which is significantly higher for V-LNMO compared to P-LNMO.

The electronic conductivity of the as-prepared slurry-coated electrodes was experimentally measured by using a collinear 4-probe set up. The conductivity of the P-LNMO electrode was ~0.128 S cm^{-1}. On the other hand, the V-LNMO electrodes exhibited a conductivity of ~0.338 S.cm^{-1} – about 2.6 times higher. To further evaluate charge transfer kinetics and the interphase, electrochemical impedance spectroscopy (EIS) was performed in a half-cell vs. Li metal after the initial formation cycle. The charge transfer resistance (R_{ct}) obtained for the P-LNMO electrode (~346.6 Ω) is ~3.26 times larger than the R_{ct} for V-LNMO (~106.3 Ω). R_{ct} has both electronic and ionic contributions and facile electron migration and Li^+ diffusion for V-LNMO is reflected in the obtained resistance values. The solid-state diffusion coefficient of Li^+ in P-LNMO and V-LNMO was calculated by linearly fitting the low frequency data from EIS. The calculated Li^+ diffusion coefficient (D_{Li}) for V-LNMO (~4.55×10^{-10} cm^2 s^{-1}) is over twice that of P-LNMO (~2.22×10^{-10} cm^2 s^{-1}). The above results demonstrate the superior electrochemical kinetics aided by the near-surface oxygen vacancies in V-LNMO when compared to the baseline P-LNMO material.

Next, we investigated the "self-discharge" behavior of P-LNMO and V-LNMO electrodes in a half-cell vs Li metal. The self-discharge profiles after a full constant current charge (0.3C, cutoff ~ 5 V) followed by constant voltage charge (5 V, cutoff ~ 0.05C) are shown in **Figure 2a**. In a fully charged state, the following reactions drive the self-discharge behavior:

$$\text{Electrolyte} \rightarrow \text{Electrolyte}^+ + e^- \quad (1)$$

$$Ni_{0.5}Mn_{1.5}O_4 + e^- + Li^+ \rightarrow LiNi_{0.5}Mn_{1.5}O_4 \quad (2)$$

The mechanism in equations 1-2 is thermally activated and is spontaneous enough to completely discharge the cell within a finite duration, even at room temperature. The open circuit voltage (OCV) of P-LNMO, after the charge cycle, drops to three subsequent plateaus in just under 120 hours, indicating a high self-discharge rate catalyzed by the P-LNMO surface. On the contrary, OCV of V-LNMO after the initial drop from 5 V, remains stable around a mean of ~4.7 V with < 0.1 V drop in the potential after 150 hours (**Figure 2a**). The self-discharge current (**Figure 2b**) is also reduced by a factor of ~3 for V-LNMO as compared to P-LNMO. Ex-situ analysis of the cathodes was carried out to reveal morphological and chemical changes associated with the interphase. SEM images of P-LNMO particles before and after the self-discharge (**Figure 2c, e**) test revealed cracks and voids because of material loss. This material loss is caused by electrolyte oxidation (which increases electrolyte acidity) and accelerates transition metal dissolution and loss of active material. On the contrary, V-LNMO exhibits a smooth surface even after the self-discharge test (**Figure 2d, f**) indicating that material loss has been suppressed. Ex situ XPS was performed on both cathodes (see **Figure 2g–j**) after the self-discharge test and as expected we observe significant differences in the cathode electrolyte interface (CEI). The O1s spectra showed formation of intense peaks of lithium and alkyl carbonates associated with the CEI. Importantly, the intensity of these peaks was much lower in V-LNMO compared to P-LNMO. The C1s spectra could be de-convoluted into C-C (carbon black), C-O, C=O, ROCO2, Li2CO3, C-F and C-H (from PVDF binder). Like O1s, the C1s spectra also suggests more intense interphase products in P-LNMO relative to V-LNMO because of the higher extent of electrolyte oxidation and thus self-discharge. These results indicate that V-LNMO effectively inhibits electrolyte oxidation and alleviates self-discharge during stand-by conditions. This behavior can be explained based on our previous observation that the V-LNMO

material lies in a partially reduced state (with higher Mn^{+3}/Mn^{+4} ratio – **Figure 1i**) due to release of electrons associated with expulsion of O^{2-} anions. Hence, the ability of V-LNMO to be further reduced (Equations 1–2) by the electrolyte, is greatly suppressed as shown in **Figure 2a**.

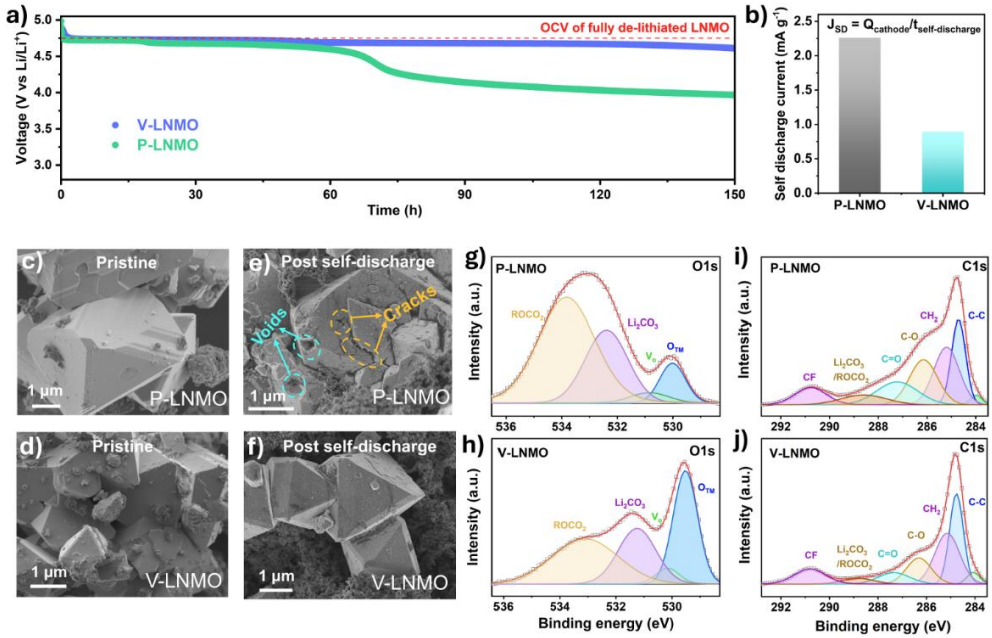


Figure 2: *(**a–b**) Potential profiles and self-discharge current of half-cells kept under open circuit voltage conditions after constant current - constant voltage charge (de-lithiation) until ~5V (vs Li/Li^+). Pristine SEM micrographs of (**c**) P-LNMO and (**d**) V-LNMO electrodes and their corresponding post-self-discharged micrographs in (**e**) and (**f**) respectively. High resolution XPS spectra for O1s for (**g**) P-LNMO and (**h**) V-LNMO and corresponding C1s spectra in (**i**) and (**j**) respectively.*

The cycling performance of P-LNMO and V-LNMO electrodes vs Li metal is shown in **Figure 3** under galvanostatic conditions and at different rates. Potential trace for both cathodes for the 1st cycle at ~0.27C (1C = 148 mA g^{-1}) is displayed in **Figure 3a**. Both the charge and the discharge halves show a higher capacity for V-LNMO (~156 mAh g^{-1} charge, ~136.9 mAh g^{-1} discharge) as compared to P-LNMO (~140 mAh g^{-1} charge, ~120.44 mAh g^{-1} discharge). Their corresponding discharge energy densities are roughly ~600 and ~560 Wh kg^{-1} for V-LNMO and P-LNMO respectively. Similarly, the potential traces at 3rd, 100th and 400th cycles at ~1.35C are compared in **Figure 3b**. Both cathodes show two prominent redox plateaus (Ni^{2+}/Ni^{4+}, Mn^{3+}/Mn^{4+}) consistent with the literature. The obtained discharge capacities for P-LNMO drastically drop from 121 → 108 → 66 mAh g^{-1} at 3rd, 100th and 400th cycles. On the other hand, V-LNMO shows discharge capacities of 129 → 121 → 117 mAh g^{-1} for these cycles. These results are a clear indication of severe irreversible parasitic reactions for the P-LNMO cathode and its much lesser impact on V-LNMO. As expected, the rate performance of V-LNMO is much better than that of P-LNMO due to its superior electronic and ionic conductivity. The discharge capacities obtained at various current densities (from ~40 mA g^{-1} to ~800 mA g^{-1}) for V-LNMO are markedly higher than that of P-LNMO (**Figure 3c**).

Once the rate performance was confirmed, long-term cycling stability tests (**Figure 3d-e**) were conducted at ~100 mA g^{-1} (0.675C), ~200 mA g^{-1} (1.35C) and ~400 mA g^{-1} (2.7C). The cycling at ~100 mA g^{-1} demonstrated a higher capacity even after 500 cycles (~131 mAh g^{-1}) for V-LNMO as compared to P-LNMO that shows a lower capacity of ~125 mAh g^{-1}. The capacity improvement can be attributed to faster kinetics and the reduced tendency of V-LNMO to oxidize the electrolyte. At higher rates of ~200 and ~400 mA g^{-1}, the differences between V-LNMO and P-LNMO are magnified greatly. P-LNMO decays severely when cycled at high rates (**Figure 3d**). Notably, at these fast rates, V-LNMO remains far more stable. For instance, the average capacity fade rate for the V-LNMO electrode over 3,000 cycles of charge and discharge is ~0.0167% per cycle (**Figure 3d**) compared to an order of magnitude higher fade rate of ~0.11% (over 600 cycles) for P-LNMO. Compared to P-LNMO, V-LNMO has a lower tendency to oxidize the electrolyte at high voltage, as previously observed in self-discharge results (**Figure 2**). Besides high voltage, interface decay is also driven by temperature rise during high-rate operation. Due to its high surface conductivity and improved kinetics, the corresponding polarizations are reduced for V-LNMO, which lowers heat generation and mitigates temperature rise at the electrode-electrolyte interface. This suppresses electrolyte oxidation and decomposition, enabling a drastic improvement in cycling stability for V-

LNMO vis-à-vis P-LNMO, especially at elevated current densities. This improvement in interface stability is apparent from the coulombic efficiency data in **Figure 3e**. P-LNMO exhibits a fluctuating pattern which periodically exceeds 100%, due to higher extent of electrolyte oxidation and induced side reactions. V-LNMO exhibits far more stable coulombic efficiency (>98%) even after long cycling (> 3000 cycles) owing to its reversibility in lithiation-delithiation reaction and less sluggish kinetics.

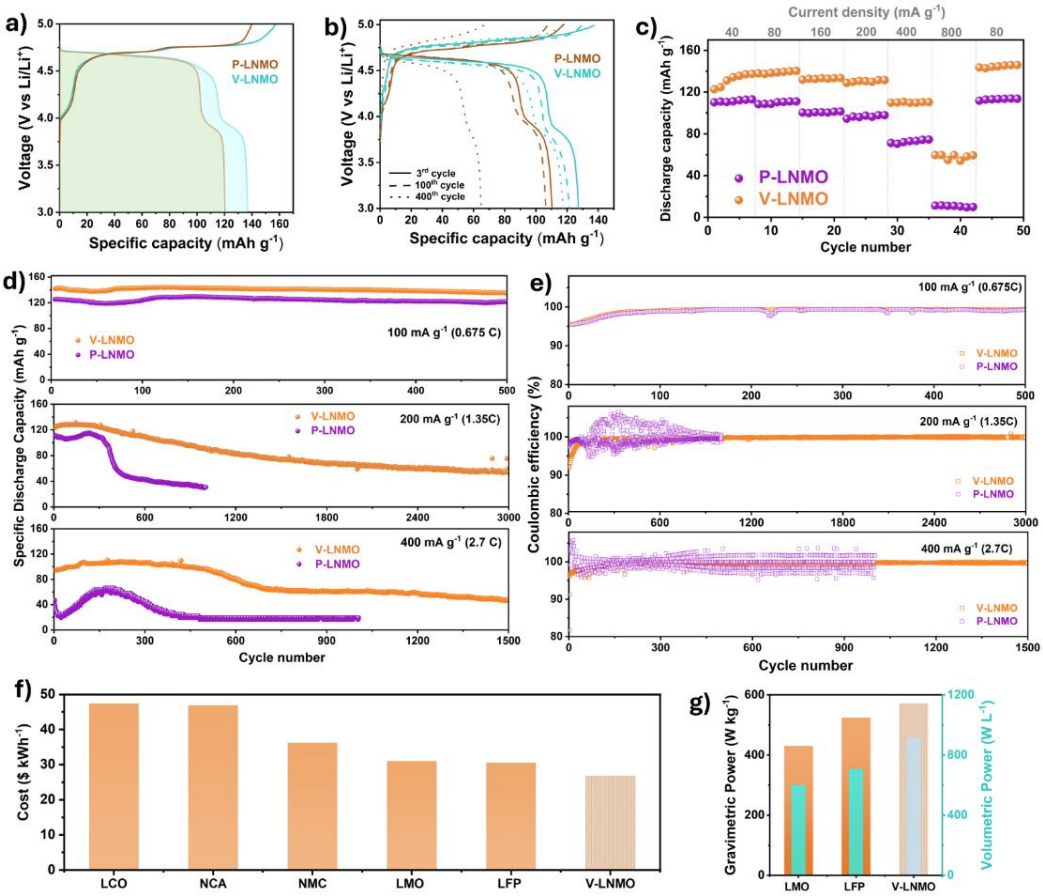


Figure 3: *(**a**) Potential trace for the 1st cycle at ~40 mA g^{-1} (~0.27C) for P-LNMO and V-LNMO and their respective discharge regions highlighting the energy output. (**b**) Potential traces at increasing cycle numbers from 3rd to 100th to 400th at an operating current density of ~200 mA g^{-1} (~1.35C). (**c**) Rate performance of both cells at different rates (~40 mA g^{-1} to ~800 mA g^{-1}) and then back to ~80 mA g^{-1}. Long term cycling stability at (**d**) ~100, ~200 and ~400 mA g^{-1} and corresponding coulombic efficiencies in (**e**). (**f**) Cost comparison in $/kWh of various commercial cathodes with V-LNMO (shaded). (**g**) Gravimetric (orange) and volumetric (cyan) power comparison of high-rate capable cathodes including V-LNMO (shaded).*

To highlight the practical significance of V-LNMO, a rough comparison of cost ($/kWh) was carried out with commercial cathode alternatives (**Figure 3f**). The cost per kWh is computed based on individual prices of the cathode materials ($/kg) divided by their respective energy density (kWh kg^{-1}).[22] The comparison is carried out at a C-rate of ~0.2C. For layered cathodes, the values are adopted from literature and for LMO, LFP and V-LNMO, in-house half-cell data was used for the evaluation. The high energy cathodes like LCO, NMC, and NCA have high cost associated with them, making each energy unit more expensive at roughly ~$47.43, ~$40.32 and ~$46.94 per kWh, respectively. By contrast, V-LNMO, LMO and LFP, are superior from the price point of view with close to ~$26.77, ~$31.04 and ~$30.59 per kWh. Our current cost estimate for V-LNMO does not include the expense associated with ammoniacal treatment and annealing. However, given the abundance of ammonia as a low-cost nitrogen source in fertilizers and nitridation processes,[23-24] the cost advantage for V-LNMO is likely to be comparable to LMO and LFP, but this has yet to be determined and will be established as part of this project.

Power density of V-LNMO was compared with two well-known high-rate capable cathodes – LFP and LMO. For comparison of the candidates, the C-rate used is ~1C. Given the high operating voltage and better compaction density of V-LNMO as compared to LFP and LMO, its power density values (gravimetric ~572 W/kg and volumetric ~915.5 W/L) are superior (**Figure 3g**) within their half-cell counterparts. The cost effectiveness of V-LNMO combined with better rate capability and power density makes this material an attractive alternative to existing cathode technologies.

2. Research Needs and Open Questions to be Addressed

While our initial results (Section 1) are promising, there are various practical issues that need to be addressed before the proposed materials can be commercialized. These include:

- We have no knowledge regarding the optimal surface oxygen vacancy concentration. Too low a concentration will result in small changes in the electronic and ionic conductivity, while too high a vacancy concentration could result in structural instability and accelerated Mn dissolution.
- It is crucially important to demonstrate that the surface oxygen vacancy concentration can be well-controlled by tuning various parameters associated with the ammoniacal treatment. This is essential to engineer V-LNMO with the optimal (desired) surface oxygen vacancy concentration.
- In our initial work, we tested V-LNMO at electrode mass loadings of 8 to 12 mg cm^{-2}. It is necessary to demonstrate feasibility at industrial standard mass loadings of 20 to 30 mg cm^{-2}.
- All data presented to date is in the half-cell format with Li metal as the counter-electrode. To demonstrate commercial viability, it is essential to demonstrate full-cells with the optimized V-LNNO cathodes paired against commercial battery-grade graphite anodes. While coin cells are a good starting point, to prove commercial viability larger form-factor pouch cells are necessary.
- Detailed electrochemical performance, manufacturing scalability and cost analysis of the full cells is required to establish feasibility of commercialization.

3. Research Plan

The previous section outlined the research needs and various issues that need to be addressed. In this section, we present our specific plans to address these research needs.

3.1 Material Synthesis & Electrochemical Testing – Achieving the Optimal (Desired) Surface Oxygen Vacancy Concentration

The procedure for creating surface oxygen vacancies is shown in **Figure 1a**. In our process, commercial LNMO powder is homogeneously added to a ~14% ammonium hydroxide solution (~ 7M) preheated to ~60°C on a hotplate and maintained at an alkaline pH of ~12. This causes NH_4^+ cations to adsorb on the LNMO and form N-O bonds with surface oxygen atoms. The suspension is stirred vigorously to maintain homogeneity and prevent any unwanted particle settling. The adsorption process is allowed to continue for ~45 min followed by vacuum filtration using a Buchner funnel. The filtered powder is allowed to dry at ~80° C for ~4 hours. In the final step, the dried powder is transferred to an alumina crucible and fired at ~500°C in ambient for ~3 hours and then allowed to cool naturally. The majority of N groups are expelled by the annealing as NO and NO_2, thus creating an "oxygen vacancy enriched zone" in the near-surface region of the LNMO particle.

Based on the above procedure, there are a number of parameters (handles) that could be used to control the density of nitrogen functional groups that are adsorbed onto the LNMO surface, including (1) concentration of the ammonium hydroxide solution; (2) temperature at which the solution is maintained during the treatment; (3) time for which the LNMO particles are exposed to the ammonium hydroxide solution; and (4) annealing temperature, which drives nitrogen and oxygen expulsion from the surface. In this task, we propose to vary these parameters systematically and study their impact on surface oxygen vacancy concentration. The concentration of the oxygen vacancies will be quantified by using X-ray photoelectron spectroscopy as shown in our preliminary work (**Figure 1g–i**). Our goal will be to tune and regulate the aforementioned synthesis parameters to systematically vary the oxygen vacancy concentration at the particle surface. This will be followed by detailed half-cell testing in coin cells (as in **Figures 2–3**) to establish the optimal concentration of surface oxygen vacancies that delivers the best electrochemical performance.

3.2 Increasing Electrode Mass Loading to Industrial Standards

In our preliminary work, we tested V-LNMO at electrode mass loadings of 8 to 12 mg cm^{-2}. The slurry making process involved mixing ~80% of cathode active powder with ~10% Super P conducting carbon additive and ~10% PVDF binder in N-methyl pyrrolidone (NMP) solvent at ~2000 RPM for ~30 min using a planetary mixer. The resulting slurry was then cast over aluminium foil current collectors using a doctor blade deposition apparatus.

In this task, we will demonstrate concept feasibility at industrial standard mass loadings of 20 to 30 mg cm^{-2}. To accomplish this, we propose to explore the following avenues: (1) Increase the slurry quantity, (2) Increase the blade gap associated with our doctor blade coating apparatus and (3) Pursue pressure-based calendaring to create a compact and well adhered coating. We expect that by tuning the above slurry-coating deposition parameters, we can produce V-LNMO electrodes at industrial standard mass loadings.

3.3 Full-Cell Testing

All data presented in our initial work was in half-cell format with Li metal as counter-electrode. To demonstrate commercial viability, it is essential to demonstrate full cell devices with the optimized V-LNNO cathodes paired against commercial battery-grade graphite anodes. Successful full cell testing will require proper capacity matching between the anode and cathode. To achieve this, we will adjust the area of the electrodes to achieve an N/P ratio (ratio of anode capacity to cathode capacity) of ~1.1. Maintaining an N/P ratio > 1, ensures that Li ions from the cathode can be fully accommodated by the anode, without the risk of Li plating or dendritic outgrowth.

Initial full cell work will be carried out at the coin cells level. Subsequently, we will build larger format pouch cells at the Center for Future Energy Systems (CFES) at RPI. These pouch cells will provide a **Minimum Viable Prototype (MVP)** that can be used for electrochemical performance (as in **Figures 2–3**) and economic (cost) analysis. **Guidance from our Korean company partner will be crucial for the proposed pouch cell MVP demonstration.**

3.4 Manufacturing and Economic Feasibility Study

The results from Tasks 3.1-3.3, will enable our team to evaluate the manufacturing scalability and economic viability of V-LNMO cathodes for large-scale industrial deployment. **Guidance from our Korean company partner will be vital for this task.**

The proposed **manufacturing scalability assessment** will include the following:

- Analyze scalability of ammoniacal treatment and annealing for high-volume V-LNMO production.
- Evaluate equipment requirements, throughput capabilities, and integration with existing battery manufacturing lines.
- Identify technical bottlenecks and recommend process modifications for industrial deployment.

The proposed **cost analysis and economic modeling** will include the following:

- Develop cost models including capital expenditure and operational expenditure for V-LNMO manufacturing.
- Perform sensitivity analysis to identify key cost drivers and their impact on total cost.
- Compare cost metrics (e.g., cost per Kg, cost per kWh) against conventional cathodes.

3.5. Milestones and Deliverables

The project timelines are summarized in the Table below.

Task	Topic	Year-1	Year-2	Year-3
3.1	Materials synthesis and testing to establish optimal surface oxygen vacancy concentration	----------		
3.2	Increasing electrode mass loading to meet industrial standards		----------	
3.3	Full cell testing in coin and then pouch cell formats			----------
3.4	Manufacturing scalability and cost analysis			----------
	Interaction with Korean partner	----------		

The **key deliverable** for this project is a fully optimized V-LNMO material, which offers the best possible combination of gravimetric and volumetric energy and power density, high-rate capability, cycle life, reduced cost and enhanced safety. Furthermore, a **minimum viable prototype (MVP)** full cell device in a pouch cell form factor will be demonstrated.

A **comprehensive final report** will also be provided to the Korean company partner that includes materials synthesis and electrode manufacturing protocols, electrochemical performance evaluation results and a detailed analysis of manufacturing scalability and economic viability. The report will also include risk assessment findings and key recommendations for industrial implementation.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The principal investigator (Prof. Nikhil Koratkar) brings expertise in battery materials and electro-chemistry. He has worked in the battery space for over 15 years and has published over 260 journal papers with > 41,000 citations (H-index = 96). Prof. Koratkar is a Fellow of the ASME, RSC, AAAS and APS. One Ph.D. student trained in battery materials synthesis and electrochemical testing will work with Prof. Koratkar on this project. Prof. Koratkar will be responsible for coordinating the research project. He will also provide regular feedback to our Korean partner company to ensure that any course corrections (if needed) can be made in a timely manner. All the required material synthesis, characterization and electrochemical testing facilities are available to Professor Koratkar on the RPI campus.
Expected R&R of Korean SME Partner	The Korean company partner would advise and assist the RPI team on two critical issues: ➔ Pouch cell testing to demonstrate a minimum viable prototype (MVP) full-cell device. ➔ Manufacturing scalability and cost analysis to establish commercial viability of the proposed technology.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	There is **no prior report** of using **surface oxygen vacancies** to boost the electronic and ionic conductivity and the interface stability of cathode materials such as spinel $LiNi_{0.5}Mn_{1.5}O_4$ (LNMO). The novel V-LNMO materials proposed in this project are promising for next-generation batteries combining the benefits of high energy given its high operating voltage (> 4.6 V), low cost (cobalt-free), fast lithium-ion transport within its 3D channels, and good safety. **The proposed project is expected to be <u>transformative</u> as it addresses the following critical needs for the battery industry: (1) high voltage operation and improved gravimetric and volumetric energy density; (2) faster c-rate charge acceptance; (3) higher power density; (4) lower cost and (5) enhanced safety. The V-LNMO cathodes that we are proposing offer a superior alternative to existing LIB cathodes, combining the high rate, cobalt-free and low-cost advantages of LFP with the high energy density of NMC and NCA cathodes.**

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization